

Size Estimation



Estimate based upon what info?

Statistics

- Size – based upon structure
- Value – based upon data

Use statistics to estimate condition selectivity

Size statistics per table

$T(R)$ or $\|R\|$	#tuples
$B(R)$	#blocks
$len(R)$	#bytes per tuple
$bf(R)$	#tuples that fit into a block (*blocking factor*)

Ignoring variable-length types, like TEXT.

If no fragmentation, $B(R) = \left\lceil \frac{T(R)}{bf(R)} \right\rceil$.

Size estimates

$\lvert R \times S \rvert$	Cross		
$\lvert R \bowtie_c S \rvert$	Inner		If joining on FK from R to S.
			Otherwise
$\lvert R ⟕_c S \rvert$	Outer		If joining on FK from R to S.
			Otherwise
$\lvert R ⋈ _c S \rvert$	Left		If joining on FK from R to S.
			Otherwise
$\lvert R \triangleright_c S \rvert$	Anti		

Size estimates

$\lvert R \cup S \rvert$	
$\lvert R \cap S \rvert$	
$\lvert R - S \rvert$	

Block size of joined tables

$T(R \times S)$	$= T(R) \cdot T(S)$	
$T(R \bowtie_c S)$	$\leq T(R)$	If joining on FK from R to S.

Recall:

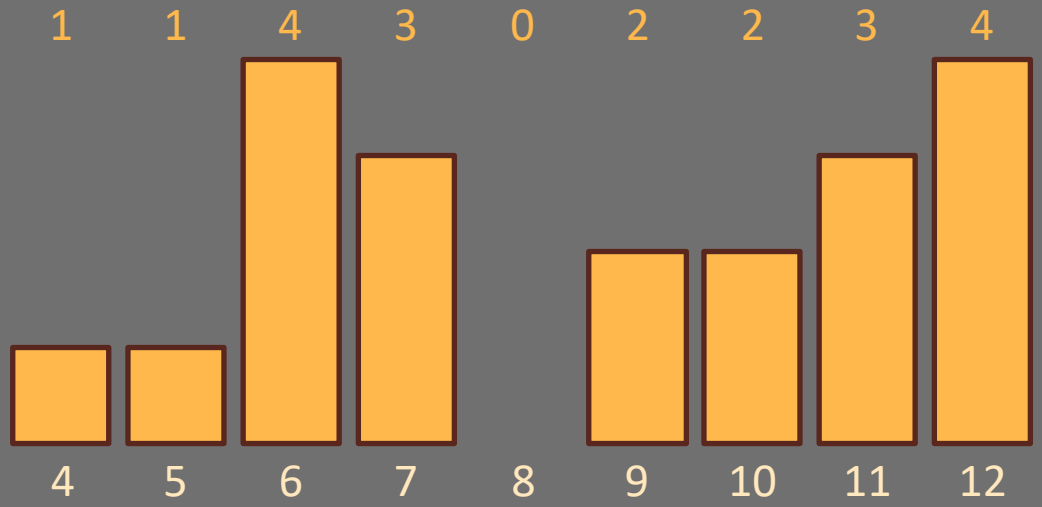
- $len(X) \cdot T(X) = B(X) \cdot bs$
- $len(XY) = len(X) + len(Y)$

$B(XY) = ?$

Value statistics for attribute R.a

$values(a)$	#distinct values
$minval(a)$	minimum value
$maxval(a)$	maximum value
histogram	Summarize value distribution

Histogram



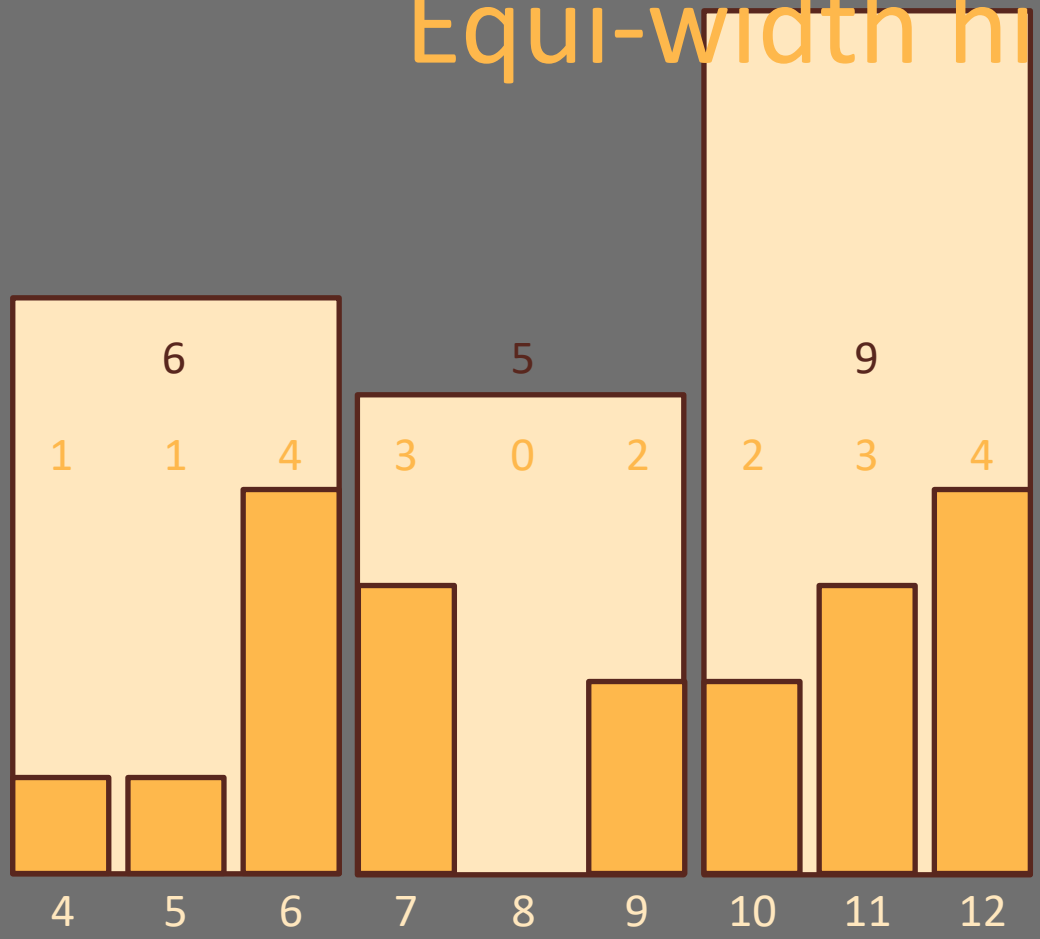
Storage space:

#values × (size(value) + size(int))

OR if values are sequential ints,

(2 + #values) × size(int)

Equi-width histogram

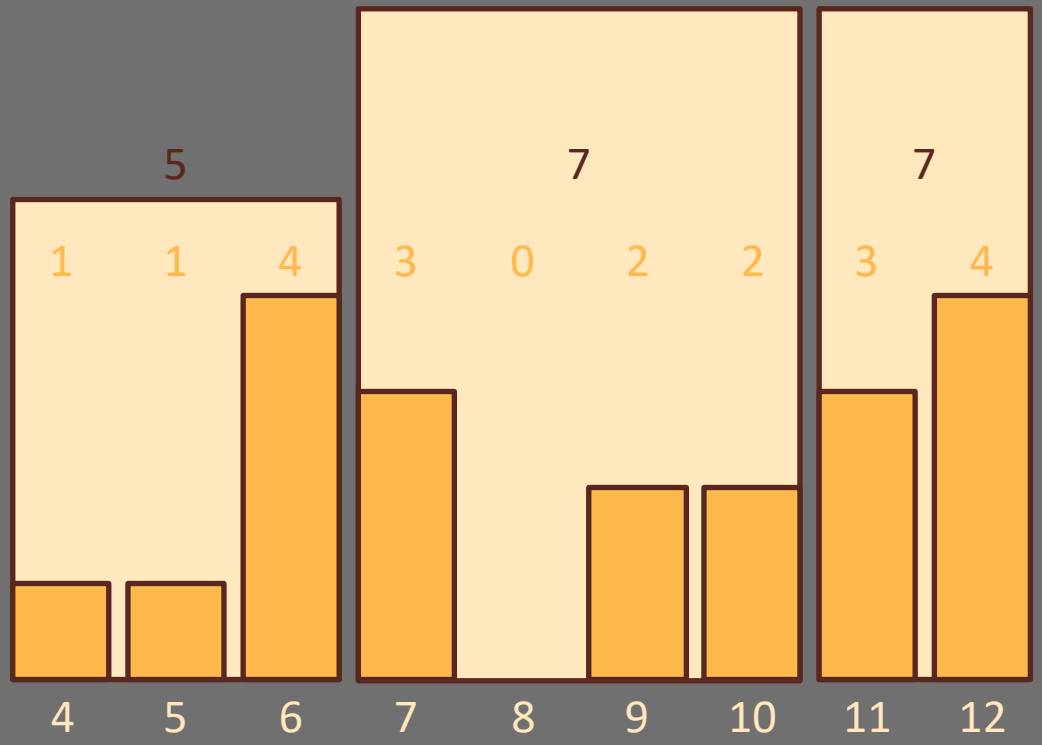


Storage space:

if values are sequential ints,

(2 + #boxes) × size(int)

Equi-depth histogram



Storage space:

if values are sequential ints,

$(2 \times \#\text{boxes} + 1) \times \text{size(int)}$

Can be more useful when distribution is non-uniform.

Histograms summary

- Several types

- Goals
  - Efficient to compute
  - Useful in estimating condition selectivity

Managing statistics

- Accuracy vs. resources
  - Update frequency
  - Histograms for which attributes? How accurate?

- Computed automatically

- Some manual control: `ANALYZE`
- Can view statistics: `SELECT * FROM pg_stats`